Healthcare Atlas

GIS and AI decision support for healthcare accessibility and future hospital planning

PROJECT REPORT FOR SUBMISSION

Prepared 10 October 2026

Submitted by: _____

Institution / Department: _____

Supervisor: _____

SYNTHETIC CHENNAI-AREA DEMONSTRATION

Not statewide · Not observed healthcare data

Submission note

Data scope disclaimer. The demonstration uses synthetic fictional zones around Chennai. It is not official Tamil Nadu geography or real healthcare evidence. Do not report these outputs as observed findings or policy recommendations.

Executive Summary

Healthcare Atlas is a local-first web decision-support prototype linking GIS accessibility analysis, demand forecasting, candidate-site screening and multi-objective hospital-location optimization. The system combines an Angular/Leaflet frontend, FastAPI application API, separate Python ML service, and MongoDB GeoJSON storage.

The demonstration uses a fully synthetic study area near Chennai, Tamil Nadu: 16 fictional zones, 5 synthetic hospitals, 18 road features, 5 boundary records, and 224 annual demand records. The workflow completed end-to-end on 10 October 2026.

In the three-hospital scenario, modeled mean travel time changed from 8.73 to 6.67 minutes and the mean accessibility index changed from 0.867 to 0.905. These are results on synthetic inputs only—not observed impacts, statewide estimates, or evidence of real-world model accuracy.

1. Introduction and Problem Statement

Healthcare access is shaped by the spatial distribution of population and facilities, road-network connectivity, travel time, capacity, and emergency-service availability. Facility counts alone cannot describe how access varies across communities or how alternative locations may change travel burdens.

This project integrates spatial data management and analysis with demand modeling and location optimization. It is intended to support exploration and comparison by analysts and planners—not to make autonomous siting or clinical decisions.

2. Aim and Objectives

Aim: demonstrate an auditable GIS and AI workflow for measuring healthcare accessibility and exploring future hospital locations.

Objectives: manage hospital, population, road, boundary, and demand layers; compute configurable distance- and road-based accessibility measures; train and evaluate demand models with chronological splits; forecast demand; screen candidate locations by population, demand, road access, separation, and land-use rules; optimize proposed sites and compare modeled before/after results; and provide interactive maps, analysis history, and exports.

3. Scope and Dataset

The live demonstration is small and synthetic. It contains 16 fictional zones around Chennai, 5 synthetic existing facilities, 18 synthetic road features, 5 boundary/land-use records, and 224 historical demand entries for 2012–2025. The modeled population totals 260,500.

This dataset is not official Tamil Nadu administrative geography and does not represent all of Tamil Nadu. Coordinates, facilities, demographics, demand, land-use boundaries, and road speeds are constructed for demonstration. No patient-level records are used. These outputs must not be cited as observed local evidence.

4. System Architecture

The Angular/TypeScript client presents dashboards, maps, and workflow controls. The FastAPI business API owns authentication, authorization, validation, data access, audit logging, and job lifecycle. MongoDB stores application

records, GeoJSON layers, workflow runs, and results. A separate FastAPI ML service performs training, prediction, and genetic-algorithm optimization. The browser does not access MongoDB directly.

Long-running operations are persisted as QUEUED, RUNNING, COMPLETED, FAILED, or CANCELLED jobs. MongoDB 2dsphere indexes support spatial queries; road travel times use a graph built from road features.

5. Methodology

Accessibility: estimate distance or road-network travel time from each population zone to facilities; combine distance, travel time, capacity, and emergency availability with configurable weights; and summarize threshold coverage and underserved population. Unreachable routes remain explicit; a documented penalty is applied only inside optimizer fitness.

Demand modeling: build lagged area/year rows using previous-period demand and demographic/spatial features. Split years chronologically: train 2013–2019, validate 2020–2022, and test 2023–2025. Compare linear regression, random forest, and XGBoost; select by validation RMSE; then recursively forecast 2027.

Candidate generation: evaluate grid points using population and predicted-demand thresholds, separation from existing hospitals, road access, study-boundary inclusion, and water/protected-area exclusions. Without parcel prices, sites receive equal relative assumed cost—not actual budget estimates.

Optimization: a genetic algorithm selects the requested number of candidates using weighted population coverage, demand coverage, accessibility improvement, road access, and cost. The system recalculates access after adding proposed facilities and reports recommendations and before/after measures. It does not guarantee a global optimum.

6. Results

Baseline accessibility: 260,500 synthetic residents; 100.0% within the configured 30-minute threshold; mean travel time 8.73 minutes; underserved population 0; mean accessibility index 0.867. Distribution: 13 Good, 3 Moderate, 0 Poor, and 0 Critical zones.

Model comparison: 208 lagged rows; linear_regression selected by validation RMSE. Held-out synthetic-test metrics: MAE 47.40, RMSE 58.47, R^2 0.994, MAPE 2.18%. These describe generated data only.

Forecast and candidate sites: the 2027 forecast estimated 36,292 synthetic demand units across 16 zones. Candidate generation returned 60 feasible points. Points are not surveyed or verified parcels.

Three-hospital scenario: mean travel time changed from 8.73 to 6.67 minutes, a 23.6% modeled reduction. Mean index changed from 0.867 to 0.905. Coverage remained 100.0% at 30 minutes; the demo therefore shows a change in mean time/index, not threshold coverage. Relative cost is 3 assumed units, at equal cost per proposed site.

7. Results Tables

Indicator	Baseline	After +3	Interpretation
Population coverage	100.0%	100.0%	Within 30 min
Mean travel time	8.73 min	6.67 min	Modeled
Mean access index	0.867	0.905	0–1 score
Relative cost	0 units	3 units	Assumed cost/site

Model	MAE	RMSE	R^2	MAPE
Linear regression	47.40	58.47	0.994	2.18%

Model	MAE	RMSE	R^2	MAPE
Random forest	91.77	133.02	0.972	3.94%
XGBoost	100.21	132.43	0.972	4.47%

Synthetic values do not indicate real-world predictive accuracy.

Tables compare the existing synthetic baseline with the three-site scenario and report the held-out temporal model comparison. Synthetic results are for software demonstration only.

8. Implementation and Validation

During preparation, the frontend production build succeeded and all three frontend tests passed. Frontend, API, and ML health endpoints responded successfully. Authenticated API checks confirmed synthetic data load, completion of each workflow stage, persistence of analysis history, and successful PDF export.

The local stack consists of the Angular development server, FastAPI API, ML service, and MongoDB. Browser end-to-end automation was not run as part of this report-preparation validation.

9. Limitations and Responsible Use

All measured values come from deterministic synthetic data. Strong test metrics do not establish performance on real populations. The 100% baseline threshold coverage reflects this small connected synthetic geography and configured threshold; it is not a Tamil Nadu access estimate.

Synthetic roads and travel assumptions omit actual traffic, public transport, seasonal access, patient choice, and referral patterns. Demand is not a clinical workload estimate. Equal site costs omit acquisition and development costs. Geometric inclusion does not establish ownership, zoning, environmental clearance, engineering suitability, community acceptance, or operational viability.

The optimizer is heuristic and does not prove global optimality. Real use requires licensed current aggregate data, independent data-quality review, geographic and temporal external validation, fairness and uncertainty assessment, security/privacy review, and multidisciplinary participation. Avoid patient-level personal information.

10. Future Work

Replace synthetic layers with licensed and quality-checked Tamil Nadu datasets: official boundaries, population projections, validated facility attributes, realistic road/travel conditions, and documented demand sources. Add spatial and temporal external validation, uncertainty intervals, and fairness analysis. Incorporate verified parcel, land-use, environmental, and cost data; compare objective weights and robust scenarios; expand browser-based end-to-end tests and deployment monitoring.

11. Conclusion

Healthcare Atlas demonstrates a complete software workflow connecting GIS data, accessibility analysis, synthetic demand forecasting, candidate-site screening, multi-objective optimization, map review, and reporting. The prototype is suitable for demonstrating system behavior and methodology; it is not a validated planning recommendation for Chennai or Tamil Nadu. Practical conclusions require real validated data and multidisciplinary review.

References and Project Materials

Internal project materials: doc/01-REQUIREMENTS.md; doc/02-ARCHITECTURE.md; doc/07-FRONTEND.md; doc/08-BACKEND.md; doc/10-EVALUATION.md; doc/11-SECURITY.md; README.md. Metrics were read from

persisted completed synthetic workflow records on 10 October 2026. No external population, hospital, road, or demand source was used for demo outputs.